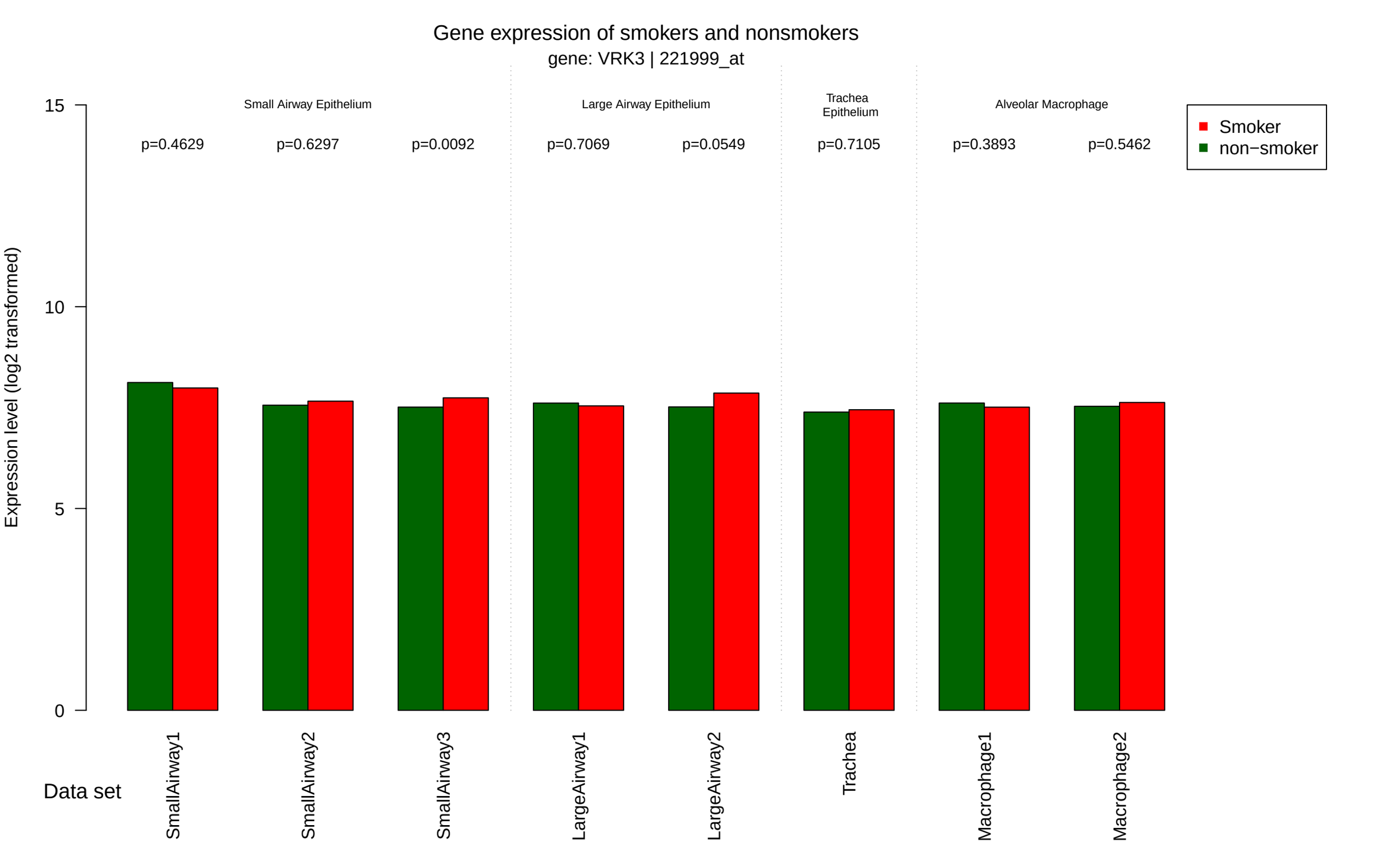
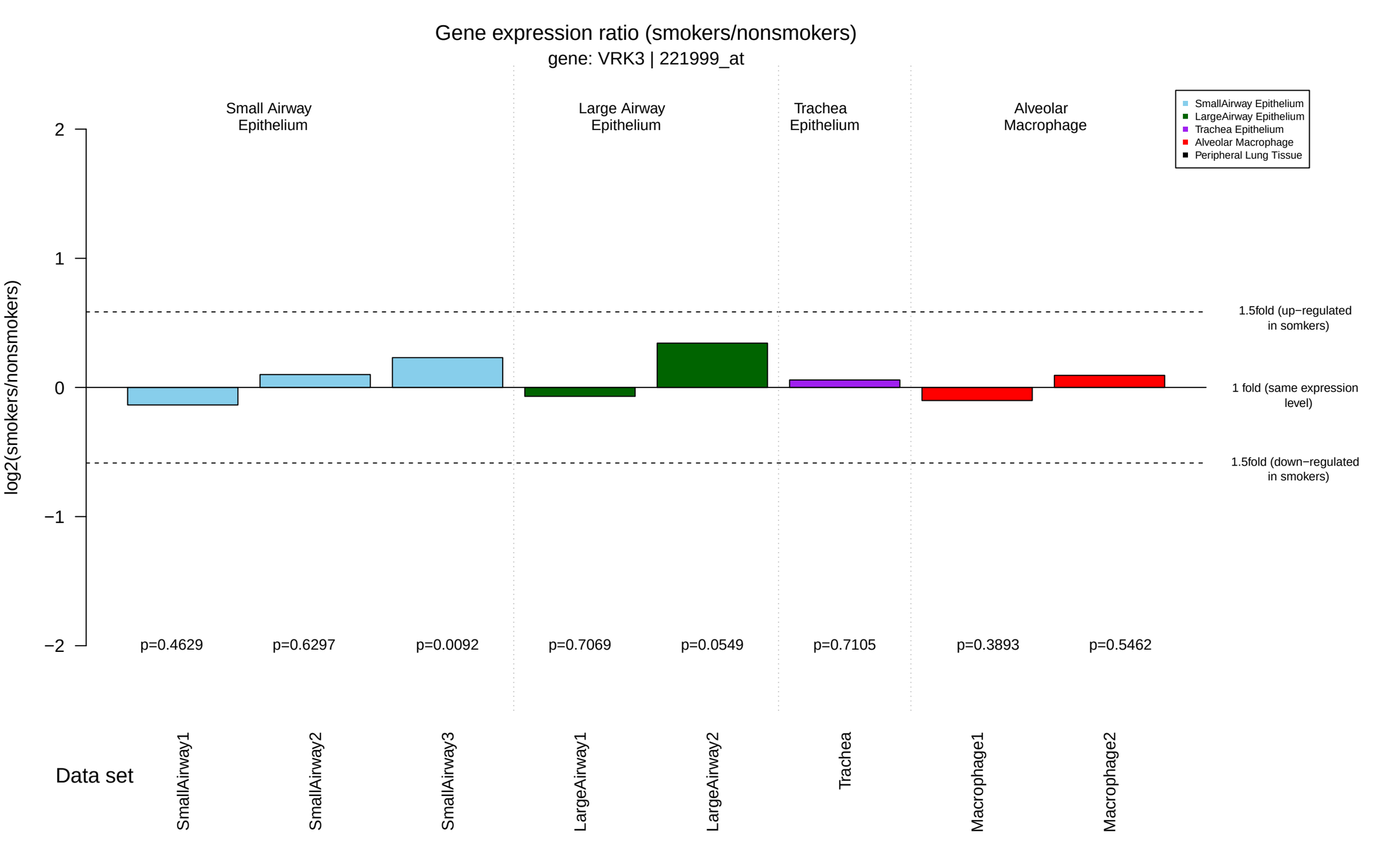
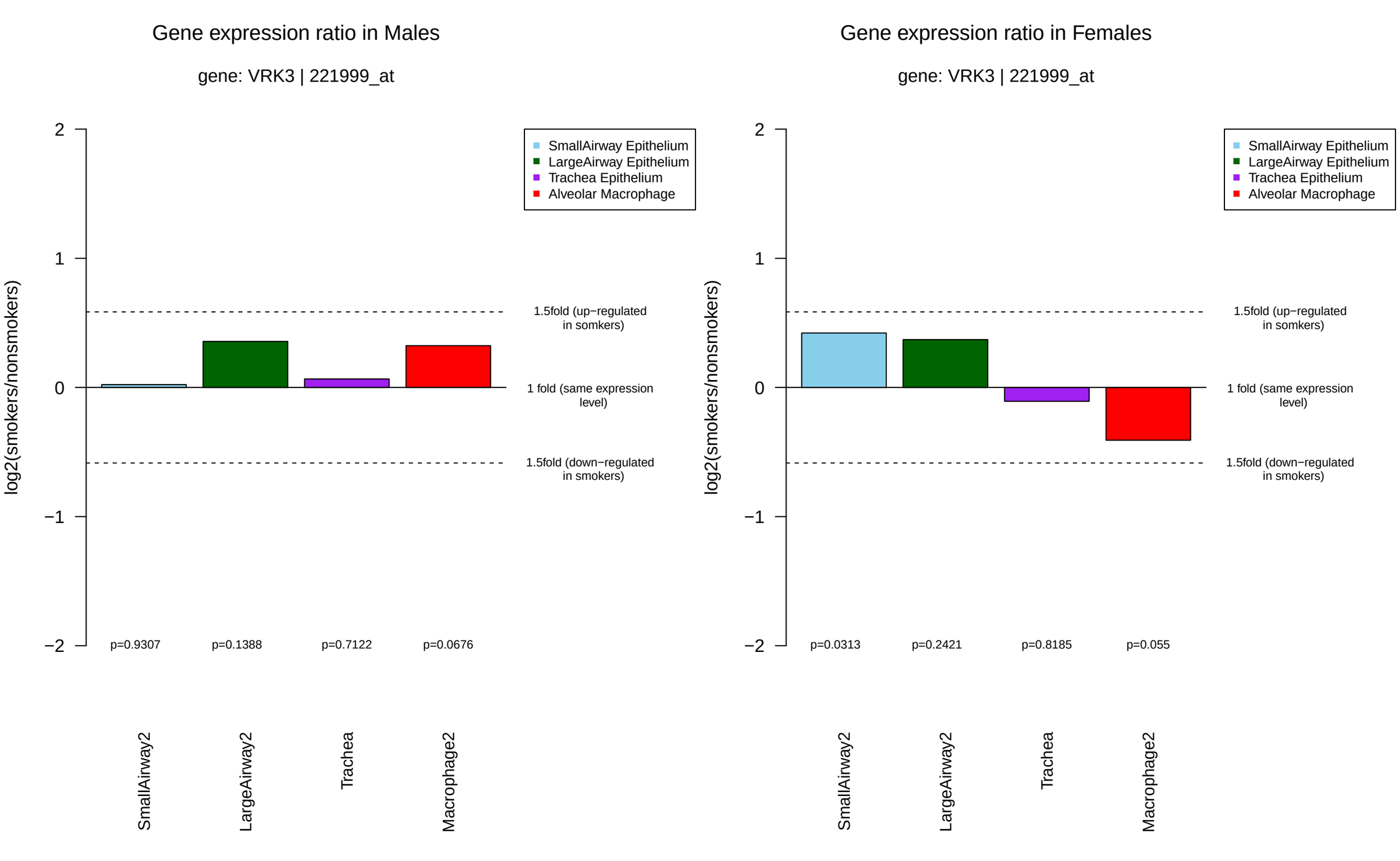
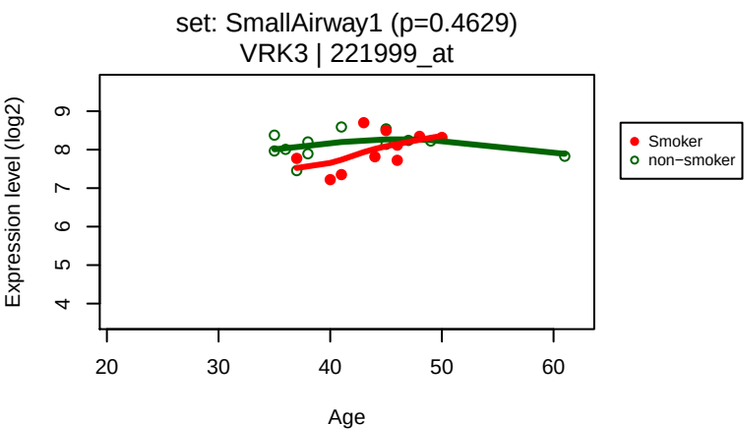
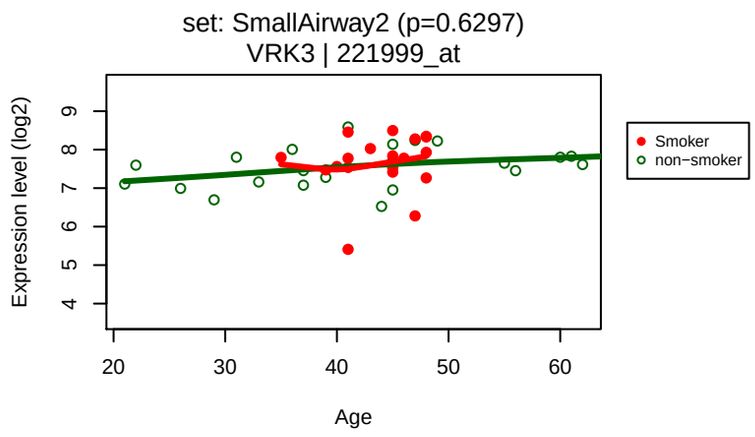
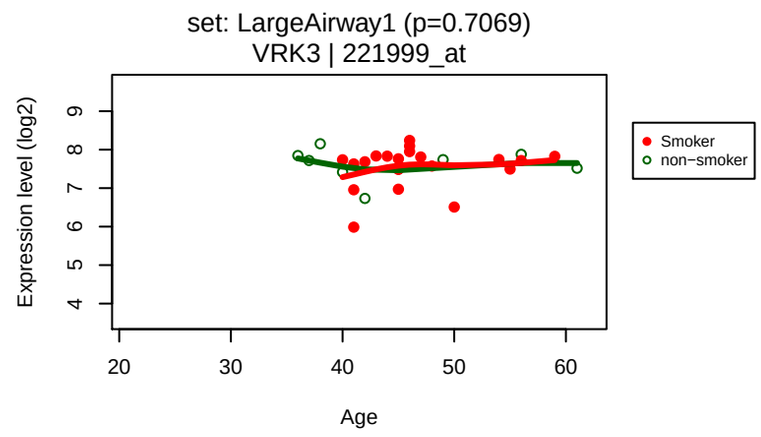

Gene expression of smokers and nonsmokers
gene: VRK3 | 221999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4629
p=0.6297
p=0.0092
p=0.7069
p=0.0549
p=0.7105
p=0.3893
p=0.5462
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

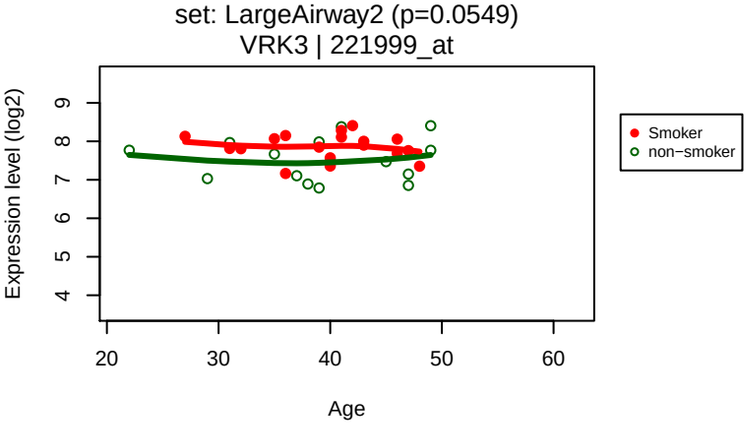

Gene expression ratio (smokers/nonsmokers)
gene: VRK3 | 221999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4629
p=0.6297
p=0.0092
p=0.7069
p=0.0549
p=0.7105
p=0.3893
p=0.5462
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

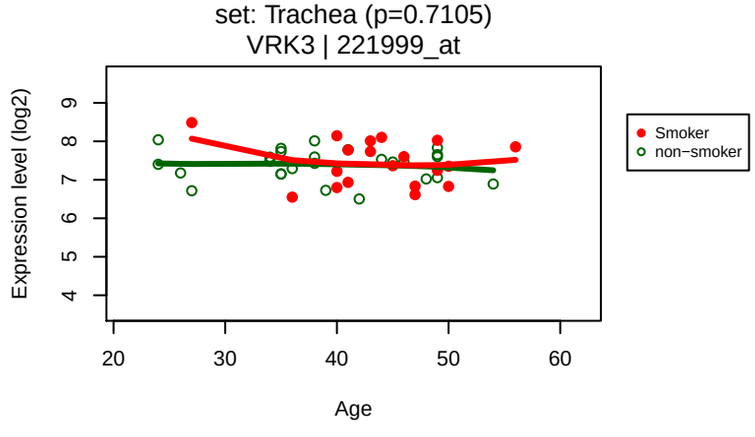

Gene expression ratio in Males
gene: VRK3 | 221999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9307
p=0.1388
p=0.7122
p=0.0676
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: VRK3 | 221999_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0313
p=0.2421
p=0.8185
p=0.055
SmallAirway2
LargeAirway2
Trachea
Macrophage2

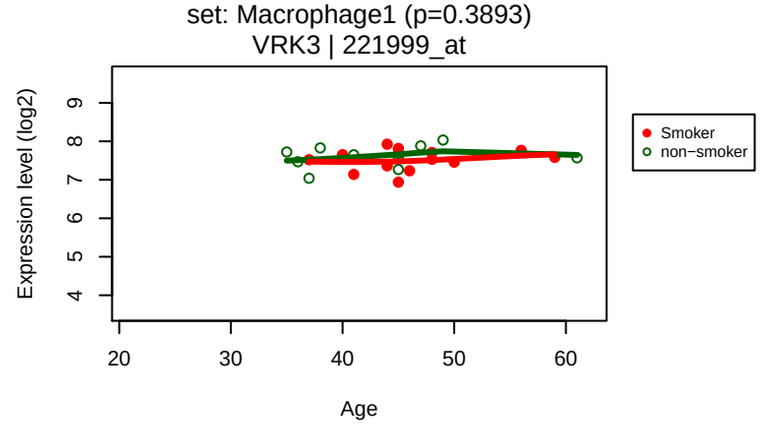

set: SmallAirway1 (p=0.4629)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker

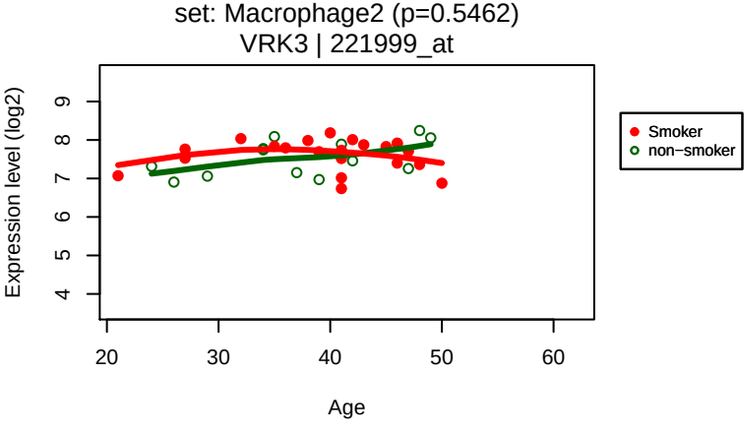

set: SmallAirway2 (p=0.6297)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker

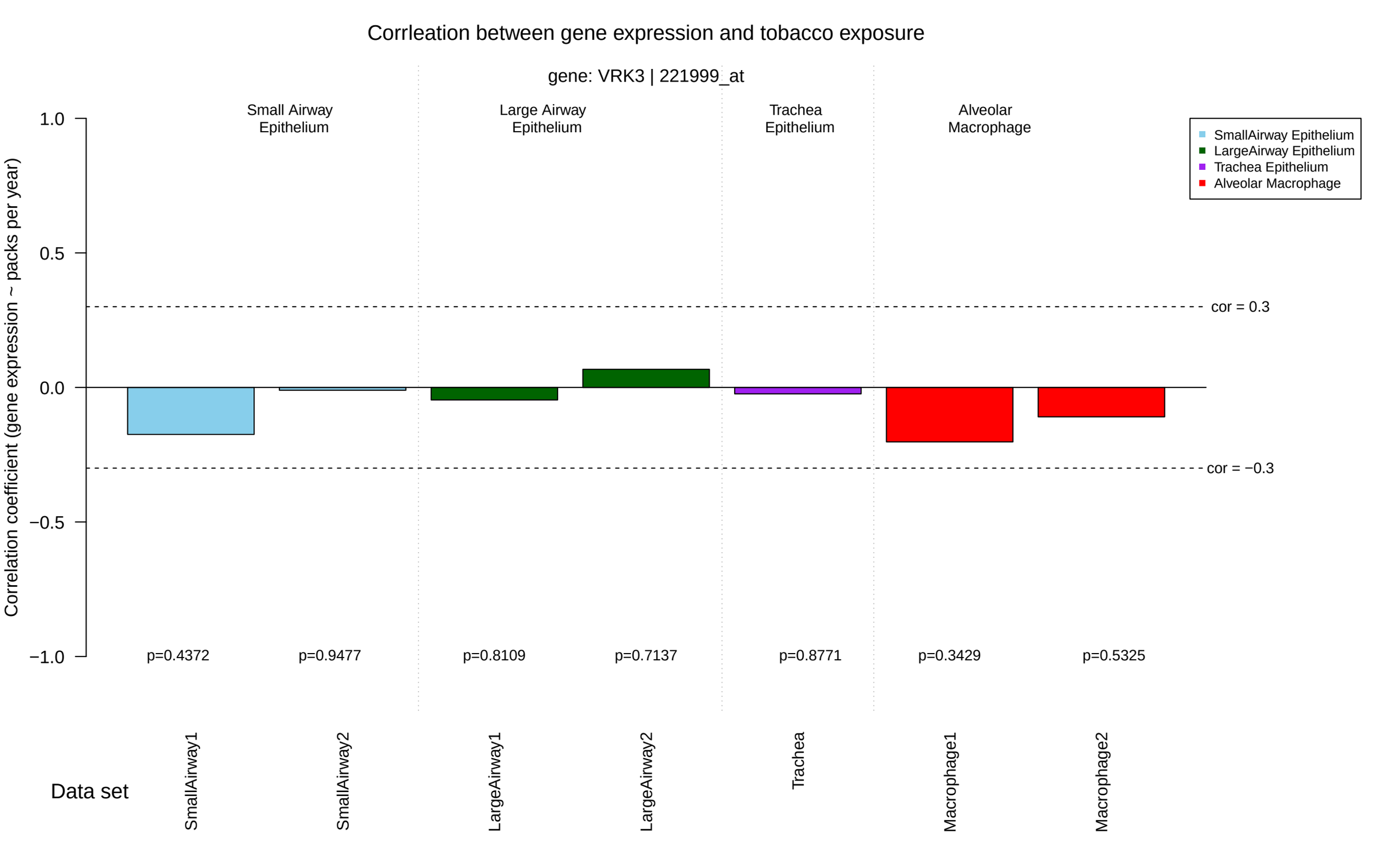

set: LargeAirway1 (p=0.7069)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0549)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7105)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3893)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5462)
VRK3 | 221999_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: VRK3 | 221999_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4372
p=0.9477
p=0.8109
p=0.7137
p=0.8771
p=0.3429
p=0.5325
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2